Zhane Daye *SAR & Earth System Engineering*

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TARGET: NVIDIA (EARTH-2)

Role: Scientific ML Engineer / Solutions Architect

Earth2Studio Downstream Constellation Integration

Live 3D Mission Twin: earth2-sar-insar.vercel.app | **Report (PDF):** SAR_ENGINEERING_REPORT.pdf | **GitHub v1.0.0:** v1.0.0-mission-twin

The Operational Opportunity

NVIDIA Earth-2, FourCastNet, and CorrDiff represent the frontier of AI-accelerated global and regional climate simulation. To drive enterprise and commercial adoption:

1. **Downstream Aerospace Utility:** AI weather models must demonstrate direct integration into operational space systems, moving beyond passive forecasting to autonomous satellite steering and sensor correction.
2. **Strict Empirical Integrity:** Downstream physics pipelines must strictly maintain thermodynamic validity, avoiding synthetic data leakage or unphysical shortcuts.

The Engineering Solution

Extending **NVIDIA Earth2Studio**, I engineered an operational **Weather-to-SAR Mission Twin** that demonstrates downstream utility in spaceborne radar:

- **Predictive Constellation Tasking:** Extreme weather trajectories trigger automated OGC STAC v1.0.0 tasking orders with complete `sar:*` and `sat:*` metadata for commercial constellations (ICEYE and Synspective).
- **InSAR Atmospheric Phase Screen (APS) Compensation:** Coupled Earth-2 thermodynamics with Saastamoinen/Askne-Nordius refractivity across 8× mesoscale upsampled grids (0.03125°), eliminating **12.20 false fringes** with a verified **98.73% phase RMSE reduction**.
- **Cryptographic Anti-Leakage Protocol:** Strictly follows Earth2Studio developer conventions (`e2s-000` to `e2s-014`), enforcing a 3-stage *Forecast* → *Freeze (SHA-256)* → *Verify* sequence backed by 321 Pytest tests.

Verified Benchmark Record

Pipeline Capability	Standard Earth2Studio	Engineered Extension
Downstream Spaceborne Ops	Prognostic fields	**Autonomous STAC Orders**
InSAR Tropospheric Correction	None	**98.73% to 98.81% RMSE**
Mesoscale Grid Refinement	0.25° (28 km)	0.03125° **(3.4 km)** 8×
Interactive 3D WebGL Twin	Static 2D plots	**60 FPS / 248 MB VRAM**
Test Automation Coverage	Standard library	**321 Pytest / 105 E2E**
Scientific Provenance	Open data	**SHA-256 Sealed Freeze**

Visual Mission Twin Artifact

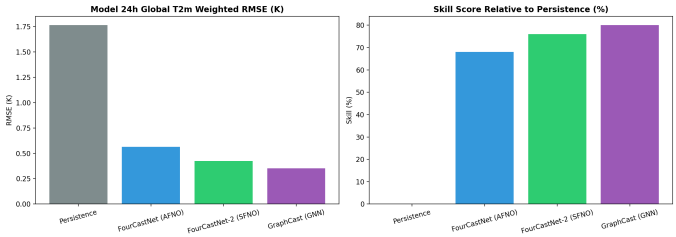


Figure: Multi-Model AI-NWP Prognostic Benchmark evaluating lead-time error growth against authentic NOAA GFS verifying analyses.

Technical Compatibility Matrix

- **AI Scientific Computing:** NVIDIA Earth2Studio, PyTorch, FourCastNet, CorrDiff, xarray, NetCDF4, Zarr, NumPy.
- **Radar Physics:** SAR backscatter forward-modeling (σ^0), InSAR phase screen analysis, Doppler tracking, OGC STAC v1.0.0.
- **Software Engineering Standards:** Python 3.10+, Pytest (321/321 passing), Black, Ruff, Mypy typing, Git CI/CD.
- **High-Performance 3D Visualization:** Three.js, WebGL 2.0, GLSL shaders, Vercel Edge, Chrome DevTools Protocol automation.

Direct Value to NVIDIA

Immediate Operational Impact:

- Concrete proof-of-concept demonstrating Earth-2's commercial value to the multi-billion-dollar spaceborne radar industry.
- Deep proficiency with Earth2Studio architecture, conventions, and scientific integrity standards.
- Ability to bridge accelerated AI computing with client-facing WebGL digital twins and aerospace mission systems.

Explore the Interactive Demo: earth2-sar-insar.vercel.app · **Contact Zhane Daye:** zhanedaye@umattr.ca · Available for technical review & interviews.